



Namibia University of Science and Technology

Topic: Municipal Financial Management System

Group Details

Steven Samuel	224034308
<u>Josenando Goncalves</u>	<u>226082628</u>
<u>Gizela Manuel(Group Leader)</u>	226029069
<u>Nataniel Antonio</u>	<u>226172015</u>
<u>Nelca Zinga</u>	<u>226046389</u>
<u>Mauipi Kasuto</u>	<u>226050750</u>
<u>Jada Shangadi</u>	224079948

Table of Content

- Introduction
- Problem Statement
- System Objectives
- System features
- Reports
- Conclusion
- Contribution List

Introduction

For this project we were assigned to a municipal Financial Management system. Municipalities handle a large amount of financial administrative information on a daily basis. Including quite important information. This includes information about employees , departmental budgets, suppliers, municipal assets and expenditure. When this information is not properly stored and organized many issues can occur and it can be difficult to keep track of records, send out proper statements , perform calculations and get hold of information when needed.

The Municipal Financial Management System was developed as a basic computerized system to help them manage not only their finances but also help manage some activities in one place. The system provides different features for employee management , budget management, supplier management , asset management and reporting. The systems users are then able to enter information , search for certain individuals records , perform calculations and view relevant information through a simple menu-driven interface.

The project development was using the C programming language and applies programming concepts

The Problem statement

Managing municipal financial and administrative information can become difficult when records are kept separately, not properly recorded or even handled manually. Information such as employee details, salaries , department budgets , supplier information and municipal assets need to be recorded and even updated regularly . Without even a simple to organize this information, users may even spend unnecessary time searching for records and performing calculations manually.

Another problem is keeping track of budgets and expenditures. If expenditures are not compared properly without the allocated budget, it will be difficult to see which department is overspending and if necessary needs my funds allocated to them and which department does not need as much funds. Similarly, supplier and assets information will near to impossible to manage when there is no central place to search and view the records from.

The problem addressed by this project is therefore the lack of a simple and organized system for managing basic municipal financial and admin information. The proposed Municipal Financial Management System aims to address this problem by providing

one menu-driven application where employee , budget , supplier and asset information can be entered , stored and searched, processed and displayed . The system also provides basic reports to make the information easier to understand and use.

System Objectives

- Allow users to add, display and search employee information
- Calculate employee salary information
- Manage department Budget
- Calculate remaining budget and Identify departments exceeding their budgets
- Add, display and search supplier information
- Maintain a basic municipal asset register
- Apply input Validation
- Demonstrate variables, operators, decisions, loops, arrays, strings and functions

System Features

- Main Menu: Navigation of all modules and Exit
- Employee Management: Add, Display, Search , Calculate Salary , Display important information
- Asset Management : Add , Display and Search for asset.
- Budget Management: Budget, expenditure, balance and Status
- Supplier Management : Add , Display and Search for a specific supplier
- Reports: Employee , budget supplier and asset reports.
- Validation: Handle invalid choices and Values.

Program Management

Program Structure

Main()

- Menu
- Employee Management :
 - addEmployee()
 - displayEmployees()
 - searchEmployee()
 - calculateSalary()
- Budget Management:
 - enterBudget()
 - calculateBudget()
 - displayBudget()
- Supplier Management:
 - addSupplier()
 - displaySupplier()
 - searchSupplier()
- Asset Management:
 - addAsset()
 - display Assets()
 - searchAsset()
- Reports:
 - displayReports()

Employee Management

Individual Contribution Record

<u>Student Name</u>	Nelca Nzinga
<u>Student Number</u>	226046389
<u>Assigned Responsibility</u>	Employee Management (Student 1)

Functions / Modules Developed

Employee Management , including:

```
#define EMPLOYEE_H
#define Max_Employees 100
struct Employee {
    int Employee_id;
    char Name [50] ;
    char Department[50] ;
    float Basic_salary ;
    float Housing_allowance ;
    float Transport_allowance ;
    int  years_of_service;
};

extern struct Employee employees[Max_Employees] ;
extern int EmployeeCount;

void Menu();

void AddEmployees(void);
void displayEmployees(void);
void searchEmployee(void);
void calculateSalary(void);
void display_Relevant_Information(void);
```

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Committed and pushed Employee.c to the main branch (commit message: "Add extern to shared variables in header "), containing the Employee Report functions listed above.

Testing Performed

```
Enter Employee_id :123
Enter Name :nelca
Department :it
Enter Basic_salary :234
Enter Housing_allowance :456
Enter Transport_allowance :345
Enter years of service :1
Employee added:
```

```
.....
      EMPLOYEE MANAGEMENT
```

```
.....
      EMPLOYEE MANAGEMENT
.....
1.Add Employee
2. Display Employee
3.search Employee
4.Calculate Salary
5. Display Relevant information
6.Exit
Enter choice
```

```
.....Display Employees.....
Employee_id : 123
Name : nelca

Department : it

Basic_salary : 234.00
Housing_allowance : 456.00
Transport_allowance : 345.00
years of service : 1
Employee_id : 24
Name : carla
```

```
Department : it

Basic_salary : 456.00
Housing_allowance : 789.00
Transport_allowance : 23.00
years of service : 2
```

```

.....Relevant Informantion.....
Employee_id : 123
Name : nelca

Department : it

Employee_id : 24
Name : carla

Department : it

```

```

Enter choice
3
Enter employee_id to search 123

Employee found
Name : nelca

Department : it

Basic_salary : 234.00
Housing_allowance : 456.00
Transport_allowance : 345.00
years of service : 1

.....
EMPLOYEE MANAGEMENT

```

Budget/ Department Management

Individual Contribution Record

<u>Student Name</u>	Nataniel António
<u>Student Number</u>	226172015
<u>Assigned Responsibility</u>	Budget Management (Student 2)

Functions / Modules Developed

Budget Management , including:

```
#define BUDGET_H
```

```
float calculateRemainingBudget(float allocatedBudget, float expenditure);
```

```
void displayBudgetStatus(float allocatedBudget, float expenditure);
```

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Committed and pushed budget.c to the main branch (commit message: "Add budget management "), containing the calculateRemainingBudget functions listed above.

Testing Performed

```
How many departments do you want to enter? 1
```

```
--- DEPARTMENT 1 ---
```

```
Enter department name: Financial
```

```
Enter allocated budget: N$-10000
```

```
Invalid! Budget cannot be negative.
```

```
Enter allocated budget again: N$10000
```

```
Enter expenditure: N$-20000
```

```
Invalid! Expenditure cannot be negative.
```

```
Enter expenditure again: N$20000
```

```
--- BUDGET INFORMATION ---
```

```
Department: Financial
```

```
Allocated Budget: N$10000.00
```

```
Expenditure: N$20000.00
```

```
Remaining Budget: N$-10000.00
```

```
Status: OVER BUDGET
```

```
--- BUDGET INFORMATION ---
```

```
Department: Financial
```

```
Allocated Budget: N$10000.00
```

```
Expenditure: N$20000.00
```

```
Remaining Budget: N$-10000.00
```

```
Status: OVER BUDGET
```

Supplier Management

Individual Contribution Record

<u>Student Name</u>	<u>Mauipi Kasuto</u>
<u>Student Number</u>	<u>226050750</u>
<u>Assigned Responsibility</u>	<u>Supplier Management (Student 3)</u>

Functions / Modules Developed

Supplier Management , including:

```
#define SUPPLIERS_H

#define MAX_SUPPLIERS 100
#define MAX_SUPPLIERS 50
void supplierMenu(void);
void addSupplier(void);
void displaySuppliers(void);
void searchSupplier(void);
void showNameLength(void);
void compareSuppliers(void);
void displaySupplierReport(void);
int getSupplierCount(void);
void supplierMenu();
void addSupplier();
void displaySuppliers();
void searchSupplier();
void compareSuppliers();
void displaySupplierReport();
```

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Committed and pushed supplier.c to the main branch (commit message: "Reformat supplier code layout"), containing the supplier report functions listed above.

Testing Performed

```
264 void supplierMenu()  
Problems Output Debug Console Terminal Ports  
  
Enter supplier name: nico  
Enter email: nicogmai.com  
Email must have @. Enter again: nico@gmail.com  
Enter telephone number: 35  
Number too short. Enter again: 354  
Number too short. Enter again: 345667878  
Enter town: grootfontein  
Supplier added.  
  
===== SUPPLIER MANAGEMENT =====  
1. Add supplier  
2. Display suppliers  
3. Search supplier  
4. Compare suppliers  
5. Back to main menu  
Enter choice: 1  
  
--- ADD SUPPLIER ---  
Enter supplier ID: -55466  
ID must be positive. Enter again: 3445  
Enter supplier name: jose  
Enter email: jose@gmail.com  
Enter telephone number: -567  
Number too short. Enter again: 345678  
Number too short. Enter again: 2334556788  
Enter town: otavi  
Supplier added.  
  
===== SUPPLIER MANAGEMENT =====  
1. Add supplier  
2. Display suppliers  
3. Search supplier  
4. Compare suppliers  
5. Back to main menu  
Enter choice: █
```

```
< supplier.c 7 0 many  
264 void supplierMenu()  
Problems Output Debug Console Terminal Ports  
  
Supplier added.  
  
===== SUPPLIER MANAGEMENT =====  
1. Add supplier  
2. Display suppliers  
3. Search supplier  
4. Compare suppliers  
5. Back to main menu  
Enter choice: 2  
  
--- SUPPLIERS ---  
  
ID: 267  
Name: queen  
Email: queen@gmail.com  
Phone: 0813544766  
Town: gobabis  
  
===== SUPPLIER MANAGEMENT =====  
1. Add supplier  
2. Display suppliers  
3. Search supplier  
4. Compare suppliers  
5. Back to main menu  
Enter choice: 3  
  
--- SEARCH SUPPLIER ---  
1. Search by ID  
2. Search by name  
3. Search by town  
Enter choice: 2  
Enter name: queen  
  
ID: 267  
Name: queen
```

```

C:\suppliers.c> main()
264 void supplierMenu()

Problems Output Debug Console Terminal Ports

ID: 267
Name: queen
Email: queen@gmail.com
Phone: 0813544766
Town: gobabis

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: 1

--- ADD SUPPLIER ---
Enter supplier ID: 7
Enter supplier name: nico
Enter email: nicogmail.com
Email must have @. Enter again: nico@gmail.com
Enter telephone number: 35
Number too short. Enter again: 354
Number too short. Enter again: 345667878
Enter town: grootfontein
Supplier added.

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: 1

--- ADD SUPPLIER ---
Enter supplier ID: -55466

```

```

PS C:\Users\Mrs. Murangi\Documents\MFMS> gcc suppliers.c -o supplier
PS C:\Users\Mrs. Murangi\Documents\MFMS> .\supplier.exe

```

```

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: 2

--- SUPPLIERS ---
No suppliers added yet.

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: 1

--- ADD SUPPLIER ---
Enter supplier ID: 267
Enter supplier name: queen
Enter email: queen@gmail.com
Enter telephone number: 0813544766
Enter town: gobabis
Supplier added.

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers

```

Assets

Individual Contribution Record

<u>Student Name</u>	Gizela Manuel
<u>Student Number</u>	226029069
<u>Assigned Responsibility</u>	Assets Management (Student 4)

Functions / Modules Developed

Assets Management, including:

```
#include <stdio.h>
#include <string.h>

struct Asset {
    int  assetID;
    char  assetName[50];
    char  assetType[40];
    double purchaseValue;
    char  department[40];
    char  condition[45];
};
//Global variable

struct Asset assets[100];
int assetCount = 0;

// Function Declaration
void addAsset(void);
void displayAsset(void);
void searchAsset(void);
void assetMenu(void);
```

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Committed and pushed assets.c to the main branch (commit message: "MFMS_assets"), containing the Assets report functions listed above.

Testing Performed

```
PS C:\Users\gizel\.vscode\ProgrammingInPratice\PAP_project> gcc assets.c -o assets
PS C:\Users\gizel\.vscode\ProgrammingInPratice\PAP_project> .\assets
-----Asset Management Menu-----
1.Add Asset
2.Display Asset
3.Search Asset
4.Back to main Menu
Choose your option: 1
Enter Asset ID: 1
Enter Asset Name: Gizela Manuel
Enter Asset Type: Office chair
Enter Asset Purchase Value: 23000
Enter Asset Department: Human Resouce
Enter Asset Condition: Good
Asset added!
-----Asset Management Menu-----
1.Add Asset
2.Display Asset
3.Search Asset
4.Back to main Menu
Choose your option: 2
Asset ID: 1
Asset Name: Gizela Manuel
Asset Type: Office chair
Purchase Value: 23000.00
Department: Human Resouce
Condition: Good
-----Asset Management Menu-----
1.Add Asset
2.Display Asset
3.Search Asset
4.Back to main Menu
Choose your option: █
```

```
Choose your option: 2
Asset ID: 1
Asset Name: Gizela Manuel
Asset Type: Office chair
Purchase Value: 23000.00
Department: Human Resouce
Condition: Good
-----Asset Management Menu-----
1.Add Asset
2.Display Asset
3.Search Asset
4.Back to main Menu
Choose your option: 3
1. Search by ID
2. Search by Name
3. Search by Type
4. Search by Department
Choose your option: 1
Enter ID: 1
--- Asset Found ---
Asset ID: 1
Asset Name: Gizela Manuel
Asset Type: Office chair
Purchase Value: 23000.00
Department: Human Resouce
Condition: Good
-----Asset Management Menu-----
1.Add Asset
2.Display Asset
3.Search Asset
4.Back to main Menu
Choose your option: 4
Back to main menu!
PS C:\Users\gizel\.vscode\ProgrammingInPratice\PAP_project> █
```

Reports

Individual Contribution Record

<u>Student Name</u>	Stephanus Samuel
<u>Student Number</u>	224034308
<u>Assigned Responsibility</u>	Reports (Student 5)

Functions / Modules Developed

Reports Management , including:

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Committed and pushed reports.c to the main branch , containing report functions listed above.

Testing Performed

```
=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM - FULL REPORT
=====

===== EMPLOYEE REPORT =====
Total Employees: 0
No employee data available.
=====

===== BUDGET REPORT =====
Total allocated budget: N$0.00
Total expenditure: N$0.00
Remaining budget: N$0.00
Departments exceeding budget: 0
No budget data available.
=====

===== SUPPLIER REPORT =====
Total Suppliers: 0
No suppliers registered in the system.
=====

===== ASSET REPORT =====
Total Assets: 0
No municipal assets registered in the system.
=====

=====
END OF COMPREHENSIVE REPORT
=====
```

Main

Individual Contribution Record

<u>Student Name</u>	Josenando Gonçalves
<u>Student Number</u>	226082628
<u>Assigned Responsibility</u>	Reports (Student 6)

Functions / Modules Developed

Reports Management , including:

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Committed and pushed main.c to the main branch (commit message: "main.c"),, containing report functions listed above.

Testing Performed

```
=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit
Enter your choice: 1

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: 4
You need at least 2 suppliers to compare.

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: █
```

```
=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM - FULL REPORT
=====
```

```
===== EMPLOYEE REPORT =====
Total Employees: 0
No employee data available.
=====
```

```
===== BUDGET REPORT =====
Total allocated budget: N$0.00
Total expenditure: N$0.00
Remaining budget: N$0.00
Departments exceeding budget: 0
No budget data available.
=====
```

```
===== SUPPLIER REPORT =====
Total Suppliers: 0
No suppliers registered in the system.
=====
```

```
===== ASSET REPORT =====
Total Assets: 0
No municipal assets registered in the system.
=====
```

```
=====
END OF COMPREHENSIVE REPORT
=====
```

```
=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
```

1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit

Enter your choice: 6

Exiting the program.

```
PS C:\Users\josen\.vscode> .\main
```

```
=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
```

1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit

Enter your choice: 5

```
===== REPORTS MENU =====
```

1. Employee Report
2. Budget Report
3. Supplier Report
4. Asset Report
5. Comprehensive Report
6. Back to Main Menu

Enter choice: 5

```
=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit
Enter your choice: 4

-----Asset Management Menu-----
1.Add Asset
2.Display Asset
3.Search Asset
4.Back to main Menu
Choose your option: 4
Back to main menu!

=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit
Enter your choice: █
```

```
PS C:\Users\josen\.vscode> .\main

=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit
Enter your choice: 1

===== SUPPLIER MANAGEMENT =====
1. Add supplier
2. Display suppliers
3. Search supplier
4. Compare suppliers
5. Back to main menu
Enter choice: 5
Going back...

=====
MUNICIPAL FINANCIAL MANAGEMENT SYSTEM
=====
1. Supplier Management
2. Employee Management
3. Budget Management
4. Asset Management
5. Reports
6. Exit
Enter your choice: 2
```

Testing, documentation , GitHub contribution (Student 7)

Individual Contribution Record

<u>Student Name</u>	Jada Shangadi
<u>Student Number</u>	224079948
<u>Assigned Responsibility</u>	Testing, documentation , GitHub contribution (Student 7)

GitHub Contribution

Repository: 226082628-Josenando/MFMS

Conclusion

Project A's objective was to design a foundation Municipal Financial Management System in ANSI C. The software demonstrates the concepts learned during the first portion of PAP such as variables, input/output, operators, decision making, loops, arrays, strings and functions.

The finished product provides elementary options for employee, budget, supplier and asset management as well as basic reports. The project provided hands-on experience in program organization, testing, debugging and collaborative development through GitHub.

Link to github Repository : <https://github.com/226082628-Josenando/MFMS>

Contribution List

Nelca – Employee Management

Nathaniel – Budget Management

Jose – Supplier Management

Gizela – Asset Management

Reports – Steven

Queen- Functions , Integration , Validation

Jada- Testing , documentation and GitHub coordination